STA100 Final term Solved Paper (Spring 2020) 26/09/2020

Q1: Define Null set.

A set that contains no elements is called null set. It is denoted by {} or ϕ.

Q2: Differentiate between constant and variable with examples?

A characteristic that varies with an individual or an object is called a **variable**. For example, age is a variable as it varies from person to person. **Constants** do not change over time. Constants are simply written as numbers.

Q3: Find A+2B from following matrices?

$A=\begin{bmatrix}2 & 1\\ 3 & 1\end{bmatrix}$, $B=\begin{bmatrix}1 & 2\\ 2 & 3\end{bmatrix}$

Ans:

$$A+2B=\begin{bmatrix}2 & 1\\ 3 & 1\end{bmatrix}+2\begin{bmatrix}1 & 2\\ 2 & 3\end{bmatrix}$$

$$=\begin{bmatrix}2 & 1\\ 3 & 1\end{bmatrix}+\begin{bmatrix}2 & 4\\ 4 & 6\end{bmatrix}=\begin{bmatrix}4 & 5\\ 7 & 7\end{bmatrix}$$

Q4: Describe the following set builder notation: $\{x / x \in N \wedge x \leq 50\}$

Set of natural numbers which are less than equal to 50. It is same as {1,2,3,…,50}

Q5: Formulate the hypothesis in case of goodness of fit test.

Ho: The fit is good

H1: The fit is not good.

Q6: Prove that $\frac{\tan y}{\sin y}=\sec y$

$$\frac{\tan y}{\sin y}=\frac{\sin y}{\cos y}.\frac{1}{\sin y}=\frac{1}{\cos y}=\sec y$$

Q7: Calculate 98% confidence interval for $\frac{\sigma_1^2}{\sigma_2^2}$ **and**

$F_{0.01(n_1-1,n_2-1)}=5.1,\ F_{0.01(n_2-1,n1-1)}=4.8,\ S_1^2=5,\ S_2^2=3.5$

Ans:

$$98\%\,C.I=\left[\frac{S_1^2}{S_2^2}.\frac{1}{F_{0.01(n_1-1,n_2-1)}},\frac{S_1^2}{S_2^2}.F_{0.01(n_2-1,n1-1)}\right]$$

$$=\left[\frac{5}{3.5}.\frac{1}{5.1}\ ,\ \frac{5}{3.5}.4.8\right]$$

$$=\left[\frac{5}{17.85}\ ,\ \frac{24}{3.5}\right]$$

$$=[0.28\ ,\ 6.86]$$

Q8: Define statistics?

That science which enables us to draw conclusions about various phenomena on the basis of real data collected on sample-basis.

Q9: Diff. b/w discrete & continuous variable.

A **discrete variable** is one that can take only a discrete set of integers or whole numbers. For example number of person in a family. **Continuous variable** can take on any value-fractional or integral within a given interval. For example weight, height, temperature etc.

Q10: Write two names of measurement scales?

1. Nominal scale

2. Ordinal or ranking scale

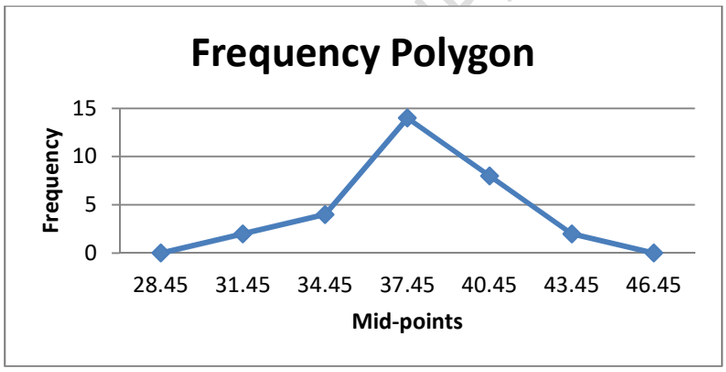
Q11: Differentiate b/w primary and secondary data?

Data that have been originally collected (raw data) and have not undergone any sort of statistical treatment, are called **PRIMARY data**. Data that have undergone any sort of treatment by statistical methods at least ONCE are called SECONDARY data.

Q12: Draw frequency polygon from the following data?

Class Boundaries	Frequency (f)	Mid-Point (X)
26.95 – 29.95	0	28.45
29.95 – 32.95	2	31.45
32.95 – 35.95	4	34.45
35.95 – 38.95	14	37.45
38.95 – 41.95	8	40.45
41.95 – 44.95	2	43.45
44.95 – 47.95	0	46.45

Frequency Polygon



Q13: Mean=7, Median=5, Standard deviation=2 find pearson coefficient of skewness & interpret the results.

$$\text{Pearson's Coefficient of Skewness} = \frac{3\,(\text{mean} - \text{median})}{\text{standard deviation}}$$

$$= \frac{3\,(7-5)}{2} \quad = \frac{3\,(2)}{2} = \frac{6}{2} = 3$$

As our answer is positive so distribution is skewed to RIGHT.

Q14: Define absolute measure of dispersion?

An absolute measure of dispersion is one that measures the dispersion in terms of the same units or in the square of units, as the units of the data.

Q15: Write three Pythagorean identities?

$$\cos^2\theta + \sin^2\theta = 1$$

$$1 + \tan^2\theta = \sec^2\theta$$

$$1 + \cot^2\theta = co\sec^2\theta$$

Q16: Write down the significance of experimental design? (2 marks)

By an experimental design, we mean a plan used to collect the data relevant to the problem under study in such a way as to provide a basis for valid and objective inference about the stated problem. The plan usually includes:

- the selection of treatments whose effects are to be studied,
- the specification of the experimental layout, and
- the assignment of treatments to the experimental units.

All these steps are accomplished before any experiment is performed.

Q17: Write formula of range.

Range=X_m-X_o

Q18: Write the formula of combined or pooled proportion of two samples.

$$\hat{p}_c = \frac{n_1\hat{p}_1 + n_2\hat{p}_2}{n_1 + n_2}$$

Q19: Use decimal parts in each value as the leaf and rest of the digits as stem to construct the stem and leaf table for the following 10 values:
7.3, 6.1, 5.2, 3.5, 4.8, 4.6, 4.4, 4.2, 5.1, 4.9

Arrange data in ascending order: 3.5, 4.2, 4.4, 4.6, 4.8, 4.9, 5.1, 5.2, 6.1, 7.3

Stem	Leaf
3	5
4	2,4,6,8,9
5	1,2
6	1
7	3

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Q20: A marble is drawn at random from a box containing 10 red, 30 white, 20 blue, 15 orange. Find the probability that drawn marble is
a. Blue
b. Red
n(S)=Total marbles=75
n(Blue)=20
n(Red)=10
P(blue)=20/75
P(Red)=10/75

Q21: Range of data is 24 and no. of classes 8. Calculate class interval.

Class interval h = RANGE/No. of classes=24/8=3

Q22: Differentiate between qualitative data and quantitative data.
Quantitative data is information about quantities. It is measureable and made with instruments such as height.
Qualitative data is descriptive. It is a phenomenon which can be observed but not measured such as hair color.

Q23: A fair coin is tossed 3 times. What is probability of at least 2 tails?
S=Sample space={ HHH,HTH,HHT,THH, TTH,THT, HTT, TTT }
n(S)=8
Let A= at least 2 tails= { TTH,THT, HTT, TTT }
P(A)=n(A)/n(S)=4/8=1/2

Q24: Define Geometric mean?

The geometric mean, G, of a set of n positive values X1, X2,…,Xn is defined as the positive nth root of their product.

Q25: Define Probability?

Probability is the measure of the likelihood that an event will occur in a Random Experiment. Symbolically we write it as P(A)=n(A)/n(S)

Q26: If A= $\begin{bmatrix} -1 & 9 \\ -8 & -2 \end{bmatrix}$ **find det[A]?**

det[A]= $\begin{vmatrix} -1 & 9 \\ -8 & -2 \end{vmatrix}$ =(-1)(-2)-(9)(-8)=2+72=74

Q27: Find Harmonic Mean if values are 10, 20, 30.

X	1/X
10	0.1
20	0.05
30	0.0333
Total	0.1833

H.M= $\frac{n}{\sum \frac{1}{X}}$ =3/0.1833=16.36

Q28: Why statistics is called science of decision making? Justify.

Statistics are useful when they are applied to improve decision making. Statistics also aid the decision making process by enabling us to establish numerical benchmarks.

Q29: Write 2 properties χ^2 of distribution.

1. It is a continuous distribution ranging from 0 to $+\infty$ .The number of degrees of freedom determines the shape of the chi-square distribution.
2. The curve of a chi-square distribution is positively skewed. The skewness decreases as *v* increases.

Q30: Write formula of mid-range?

$$\text{mid} - \text{range} = \frac{x_0 + x_m}{2}$$

Q31: Write formula of LSD?

LEAST SIGNIFICANT DIFFERENCE (LSD):

$$LSD = t_{\alpha/2,(\nu)}\sqrt{\frac{2(MSE)}{r}},$$

Q32: Write linear regression equation for y on x?

Y=a+bx, where Y is dependent variable, X is independent variable, b is slope of line and a is y-intercept.

Q33: Define correlation?

Correlation is a measure of the strength or the degree of relationship between two RANDOM variables.

Q34: A fair coin is tested what is the probability of getting number of tails?

S=Sample space={ H,T }
n(S)=2
P(No. of tails)=1/2

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Q35: How many parameters are associated with F-distribution and what is the range of the distribution?

This distribution has two parameters v_1 and v_2, which are known as the degrees of freedom of the F-distribution.

Range: The F-distribution is a continuous distribution ranging from zero to plus infinity.

Q36: Write down any two statistical software widely used for statistically sampling.

SPSS, Minitab

Q37: For the given data mean25, variable=16 & median=22. Calculate pearson's coefficient of skewness.

Standard deviation= $\sqrt{\text{var}\,iance}$ = $\sqrt{16}$ =4

Coefficient of skewness $= \dfrac{3\,(\text{mean} - \text{median})}{\text{s tan dard deviation}}$

Coefficient of skewness $= \dfrac{3\,(25-22)}{4}$

$= \dfrac{3(3)}{4}$

=9/4

Q38: A single card is drawn from a deck of 52 playing cards, what is the probability of

a. An Ace

b. A Jack

n(Ace)=4

n(Jack)=4

P(Ace)=4/52

P(Jack)=4/52

Q39: If Q1=65 and Q3=85. Find Quartile deviation (Q.D) & coefficient of Quartile deviation.

$$Q.D. = \frac{Q_3 - Q_1}{2}$$

$Q.\text{D} = \dfrac{85-65}{2}$ =20/2=10

Q40: Two sets A= $\begin{bmatrix} 2 & 3 & 4 \\ 1 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$ **and B=** $\begin{bmatrix} 2 & 4 & 9 \\ 6 & 5 & 2 \end{bmatrix}$ **are multiplied or not? Justify.**

These matrix cannot be multiplied because the number of columns in the first **matrix** is not same as the number of rows in the second **matrix.**

Q41: Find the range and coefficient of range of following values:

57, 11, 39, 56, 82, 11, 64,18,37,15

Xo=11, Xm=82

Range=Xm-Xo=82-11=71

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Coefficient= $\frac{X_m - X_0}{X_m + X_0} = \frac{82-11}{82+11} = \frac{71}{93}$

Q42: Calculate mid-points which are used for frequency polygon.

Class Boundaries	Frequency (f)	Mid-Point (X)
26.95 – 29.95	**0**	**28.45**
29.95 – 32.95	2	31.45
32.95 – 35.95	4	34.45
35.95 – 38.95	14	37.45
38.95 – 41.95	8	40.45
41.95 – 44.95	2	43.45
44.95 – 47.95	**0**	**46.45**

Q43: Average Height of the students in a school is 5.2 inches. A sample of 12 students showed the following heights.

5.0, 5.3, 5.2, 4.9, 4.11, 5.0, 5.5, 5.4, 5.1, 5.0, 5.2, 4.10

Calculated the sampling error? (5 marks)

Here $\mu = 5.2$, $\bar{X} = \frac{\sum X}{n} = \frac{59.81}{12}$ =4.98

Sampling Error= $\bar{X} - \mu$ = 4.98 - 5.2 = -0.22

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Q44: Find the center and radius of the given equation of circle

$(x+2)^2 + (y-3)^2 = 9$ (3 marks)

$(x-(-2))^2 + (y-3)^2 = 3^2$

The center (-2,3) and radius is 3

Q45: Write down the formula for finding the sum of thes first n terms of a geometric sequence. (2 marks)

$$G = \sqrt[n]{X_1 X_2 \ldots X_n}$$

Q46: Name the measures of dispersion that are NOT based on all the values. (2 marks)

Range

Quartile Deviation

Q47: Define set?

A set is an unordered collection of distinct objects. Objects in the collection are called elements of the set.

Q48: Write two disadvantages of range?

1. It ignores all the INFORMATION available from the intermediate observations.

2. It might give a MISLEADING picture of the spread in the data.

Q49: Find the mode from the given data 50, 60, 80, 100, 90, 80.

Most repeated value is 80. Mode=80

Q50: Find the geometric mean of numbers: 45, 32, 37, 46, 39, 36, 41, 48, 36.

X	log X
45	1.6532
32	1.5052
37	1.5682
46	1.6628
39	1.5911
36	1.5563
41	1.6128
48	1.6812
36	1.5563
	14.3870

$$\log G = \frac{\sum \log X}{n}$$

$$= \frac{14.3870}{9} = 1.5986$$

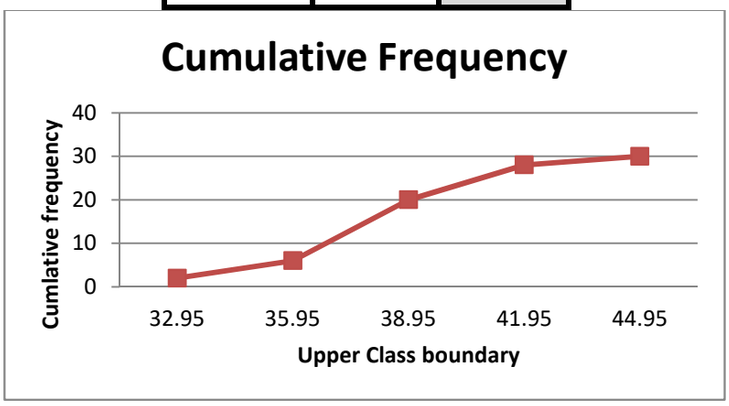
$$\text{Hence } G = \text{anti}\log 1.5986$$

$$= 39.68$$

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Q51: Draw cumulative frequency polygon from the given data.

Class Boundaries	Frequency	Cumulative Frequency
29.95 – 32.95	2	2
32.95 – 35.95	4	6
35.95 – 38.95	14	20
38.95 – 41.95	8	28
41.95 – 44.95	2	30



Q52: Question: The sample mean ($\bar{X}$) is 135 and population (μ) is 129. Calculate sampling error:

Sampling error = $\bar{X} - \mu$ =135-129=6

Q53: If $\sum X^2 = 0.729,\ \sum X = 0.701,\ \bar{X} = 0.140,\ n - 5$ then find coefficient of variation?

$$S=\sqrt{\frac{\sum X^2}{n}-\left(\frac{\sum X}{n}\right)^2}\ ,\ S=\sqrt{\frac{0.729}{5}-\left(\frac{0.701}{5}\right)^2}$$ =0.36

$$CV=\frac{S}{\bar{X}}\times 100=\frac{0.36}{0.140}\times 100$$ =253.69

Q54: Find $(5+4i)(-9-7i)$

$(5+4i)(-9-7i)$ = -45-35i-36i-28i^2 = -45-71i-28(-1) = -45-71i+28 = -17-71i

Q55: Find $z.z^*$ **when z=(3-7i)**

Z^*= (3+7i)

$z.z^*$ = (3-7i) (3+7i) = 9+21i-21i-49i^2 = 9-49(-1) = 9+49 =58

Q56: If F-value is 1.13 and error mean square is 24.13, find the value of mean square of treatment.

$$F=\frac{M.S.Treatment}{M.S.Error}$$

$$1.13=\frac{M.S.Treatment}{24.13}$$

$$M.S.Treatment=1.13\times 24.13=27.33$$

Q57: Find 95% confidence interval of normal distribution or of population mean μ **?**

the 95% C.I. for μ is given by

$$\bar{X}\pm 1.96\frac{\sigma}{\sqrt{n}}$$

VU SUPREME TALEEM

Q58: Write assumptions of chi square test of goodness of fit.

1. The total number of observations (i.e. the sample size) should be at least 50.
2. The expected number ei in any of the categories should not be less than 5.
3. The frequencies of the categories should be independent

Q59: Define P-value?

The probability of observing a sample value as extreme as, or more extreme than, the value observed, given that the null hypothesis is true.

Q60: Write χ^2 **test statistics for goodness of fir?**

$$\chi^2=\sum_i\sum_j\left(0_{ij}-e_{ij}\right)^2/e_{ij}$$

Q61: Why Venn diagrams are used in set theory?

The Venn diagrams are used to represent sets and subsets in a pictorial way and to verify the relationship among sets and subsets.

Q62: Why stem and leaf table is preferred over frequency distribution?

In identity of individual observations is lost in grouping process. That's why we use stem and leaf table.

Q63: Differentiate between ratio and interval scale.

Ratio scale is an interval scale where the scale of measurement has a true zero point as its origin. The ratio scale is used to measure weight, volume, distance, money, etc.

Interval scale is a measurement scale possessing a constant interval size (distance) but not a true zero point, for example temperature.

Q64: A fair coin is tested what is the probability of getting head?

S=Sample space={ H,T }
n(S)=2
P(head)=1/2

Q65: In how many ways can 5 people be seated on 7 chairs a round table if only their positions relative to each other count (that is, the arrangements obtained from each other by rotation of all people are considered to be the same).

7 places to seat person #1, leaving
6 places to seat person #2, leaving
5 places to seat person #3, leaving
4 places to seat person #4, leaving
3 places to seat person #5
That's 7x6x5x4x3 = 2520

Then we divide that by 5 rotations and get 504 ways.

VU SUPREME TALEEM

Q66: Find the slope of the line given 3x + 2y = 9

Writing the equation in the slope-intercept form, we get:

3x + 2y = 9
2y=9-3x
Y=$\frac{9}{2}-\frac{3x}{2}$
Comparing it with Y=a+bX where b is slope of line
Therefore, slope =-3/2

Q67: Find out mean deviation and $(X-\bar{X})=4,2,1,1,2,4$ **.**

$$M.D=\frac{\sum(X-\bar{X})}{n}=\frac{4+2+1+1+2+4}{6}=\frac{14}{6}=2.33$$

Q68: Evaluate/Solve the polynomial x^2+6x+5 Complete in square form.

$x^2+6x+5=0$

$x^2+2(x)(3)+9-9+5=0$

$(x+3)^2-4=0$

$(x+3)^2=4$

$x+3=\pm\sqrt{4}$

$x=\pm 2-3$

$x=-2-3,\ x=2-3$

$x=-5, \qquad x=-1$

Q69: If A = {dog, horse, bird, rabbit, cat} and B = {cat, rabbit} and C={lion, monkey} then find AUB and?

A ∪ B = { dog, horse, bird, rabbit, cat }

Q70: From the set A = {3, 4, 6} and B = {1, 2, 3, 5, 6} then find A ∩ B?

A ∩ B = {3, 6}

Q71: Given, n1=n2=16, S_1^2 =50, S_2^2 =16. Construct a 90% confidence interval for the variance ratio $\frac{\sigma_1^2}{\sigma_2^2}$

The 90% confidence interval for $\frac{\sigma_1^2}{\sigma_2^2}$ is given by

$$\left[\frac{S_1^2}{S_2^2}.\frac{1}{F_{0.05(n_1-1,n_2-1)}},\frac{S_1^2}{S_2^2}.F_{0.05(n_2-1,n1-1)}\right]$$

$$=\left[\frac{50}{16}.\frac{1}{F_{0.05(15,15)}},\frac{50}{16}.F_{0.05(15,15)}\right]=\left[\frac{50}{16}.\frac{1}{2.40},\frac{50}{16}.2.40\right]$$

= (1.31, 7.47)

Q72: Write a probability function f(x) normal distribution?

The probability density function is given by

$$f(x)=\frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{1}{2}\left[\frac{x-\mu}{\sigma}\right]^2}, \qquad -\infty<x<\infty$$

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Q73: Define negatively skewed curve?

The negatively skewed frequency curve is the one for which the left tail is longer than the right tail.

Q74: For a certain distribution mean is 141, mode is 122.3 and S=50. Calculate Pearson coefficient of skewness.

Mean =141, Mode=122.3, S=50

Pearson's Coefficient of Skewness = $\frac{mean - \mathrm{mod}\, e}{S.D} = \frac{141-122.3}{50} = \frac{18.7}{50} = 0.374$

Q75: A random sample of size 24 is drawn from normal population with mean=44 and σ^2 . If S=15, $\bar{X}$ =40.4 then calculate t-statistics.

$$t = \frac{\bar{X} - \mu}{s/\sqrt{n}}$$

$$t = \frac{40.4-44}{15/\sqrt{24}} = \frac{-3.6}{3.06} = 1.176$$

Q76: In normal distribution $\mu = 20$ and $\sigma = 4$. What percentage of cases will fall between 16 and 24?

Z score for 16 cases $\frac{X-\mu}{\sigma} = \frac{16-20}{4} = -\frac{4}{4} = -1$

Z score for 24 cases $\frac{X-\mu}{\sigma} = \frac{24-20}{4} = \frac{4}{4} = 1$

Cases will fall between mean and + 1σ.

Q77: A certain type of storage battery lasts on the average 3.0 years, with a standard deviation of 0.5 year. Assuming the battery lives are normally distributed, find the probability that a given battery will last less than 2.3 years.

- We want to find P(X < 2.3) to find this area or probability by computing z as

$$z = \frac{X-\mu}{\sigma} = \frac{2.3-3}{0.5} = -1.4$$

P(X < 2.3) = P(z < -1.4)
From the normal distribution table we can conclude that the probability is 0.0808

Q78: Find the probability of drawing a white ball from a bag containing 4 red, 8 black, and 3 white balls. (3 marks)

n(S)=4+8+3=15

P(White ball)=n(white balls)/n(s)

=3/15

Q79: What are the mean and variance of binomial distribution?

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The mean of the distribution μx= np.

The variance σ^2x=npq.

Q80: If we have n =15 and $\sum_{i=1}^{n}(X_i-\bar{X})^2=56$ .Then, find the 90% confidence interval for population variance (σ^2).

The 90% confidence interval for σ^2 is given by

$$\frac{\sum(\mathrm{Xi}-\bar{\mathrm{X}})^2}{\chi^2_{0.05(n-1)}}<\sigma^2<\frac{\sum(\mathrm{X_i}-\bar{\mathrm{X}})^2}{\chi^2_{0.95(n-1)}}$$

$$\frac{56}{\chi^2_{0.05(14)}}<\sigma^2<\frac{56}{\chi^2_{0.95(14)}}$$

$$\frac{56}{23.68}<\sigma^2<\frac{56}{6.57}$$

$$0.0358<\sigma^2<8.5236$$

Q81: Write a short note on quota sampling.

In Quota sampling, a sampling frame is not used at all, and the choice of the actual sample units to be interviewed is left to the discretion of the interviewer.However, the interviewer is restricted by quota controls.

Q82: Write down the testing procedure in case of goodness of fit test. (2 Marks)

HYPOTHESIS-TESTING PROCEDURE

Step-1:

H_0 : The fit is good

H_A : The fit is not good

Step-2:

Level of Significance: $\alpha = 0.05$

Step-3: Test-Statistic:

$$\chi^2=\sum_i\frac{(\mathrm{o_i}-\mathrm{e_i})^2}{\mathrm{e_i}}$$

which, if H0 is true, follows the chi-square distribution having k - 1 - r degrees of freedom(where k = No. of x-values (*after* having carried out the necessary mergers), and r = number of parameters that we estimate from the sample data).

Step-4: Computations:

Step5: Critical region

Step6: Conclusion

Q83: What does quartile deviation measure in a distribution?

The quartile deviation is defined as half of the difference between the third and first quartiles.

Q84: 62% of students get passed in a particular test. This time 500 students appear in the exam using the normal approximation to the binomial distribution. Find the mean and variance of this distribution.

n=500

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Student passed as probability of success=p=62%=0.62

Student failed as probability of failure=q=1-p=1-0.62=0.38

Mean=np=500x0.62=310

Variance= npq= 500x0.62x0.38=117.8

Q85: Find cumulative frequency from the following table

Size of shoes	No. of pairs
5	2
6	1
7	4
8	6
9	5

Ans:

Size of shoes	No. of pairs	Cumulative frequency
5	2	2
6	1	3
7	4	7
8	6	13
9	5	18
Total	**18**	

Q86: Given are the ages of first year student of a particular college. Calculate the Harmonic mean.

X	1/X
13	0.076923
14	0.071429
15	0.066667
16	0.0625
17	0.058824
Total	0.3363

H.M= $\frac{n}{\sum \frac{1}{X}}$ =5/0.3363=14.87

Q87: Multiple the given matrices.

$$\begin{bmatrix} 3 & 12 & 4 \\ 5 & 6 & 8 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 7 & 3 & 8 \\ 11 & 9 & 5 \\ 6 & 8 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 3*7+12*11+4*6 & 3*3+12*9+4*8 & 3*8+12*5+4*4 \\ 5*7+6*11+8*6 & 5*3+6*9+8*8 & 5*8+6*5+8*4 \\ 1*7+0*11+2*6 & 1*3+0*9+2*8 & 1*8+0*5+2*4 \end{bmatrix}$$

$$= \begin{bmatrix} 177 & 149 & 100 \\ 149 & 133 & 102 \\ 19 & 19 & 16 \end{bmatrix}$$

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Q89: Find the coefficient of standard deviation from the following data. Life in Hours (X): 130, 150, 180, 250, 345.

$\bar{X} = \frac{\sum X}{n}$ =1055/5=211

X	$X - \bar{X}$	$(X - \bar{X})^2$
130	**-81**	**6561**
150	**-61**	**3721**
180	**-31**	**961**
250	**39**	**1521**
345	**134**	**17956**
1055	**0**	**30720**

S= $\sqrt{\frac{\sum (X - \bar{X})^2}{n}} = \sqrt{\frac{30720}{5}} = 78.38$

Coefficient of S.D $= \frac{S}{\bar{X}} = \frac{78.38}{211} = 0.37$

Q90: For the 10 values, we are given $\sum |X_1 - \bar{X}| = 20$. Find mean deviation.

$M.D = \frac{\sum |X_1 - \bar{X}|}{n} = \frac{20}{10} = 2$

Q91: When we apply finite population correction (fpc) easily?

In sampling from a finite population, we apply the fpc only if the sample size is greater than 5% of the population size.

Q92: Make Stem and leaf table for the given data: 59, 46, 50, 54, 47, 39

Arrange data in ascending order: 39, 46, 47, 50, 54, 59

Stem	**Leaf**
3	9
4	6 7
5	0 4 9

Q93: Apply LSD test on the following data:

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ANOVA-TABLE

Source of Variation	d.f.	Sum of Squares	Mean Square	F
Between Treatments (i.e. Between Rations)	3	134.45	44.82	$F_1 = 20.47$
Between Blocks (i.e. Between Groups)	4	21.46	5.36	$F_2 = 2.45$
Error	12	26.26	2.19	--
Total	19	182.17	--	--

Solution:

$$LSD = t_{\alpha/2,(v)}\sqrt{\frac{2(MSE)}{r}}$$
$$= t_{0.025(12)}\sqrt{\frac{2(2.19)}{5}}$$
$$= 2.179\sqrt{\frac{2(2.19)}{5}}$$
$$= 2.179 \times 0.936$$
$$= 2.04.$$

Q94: Find the 10th term of the geometric sequence with a = 2000 and a common ratio of 1/2.

a10 = 2000• (1/2)9

= 2000 • 1/512

= 2000/512 = 500/128 = 250/64 = 125/32

Q95: Given the sequence, 1500, 3000, 4500, 6000, …, find the 16th term.

Solution: a = 1500, d = 1500, n -1 = 16 - 1 = 15 .

So a16 = 1500 + 1500(15) = 24000

Q96: Write Quadratic formula?

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Q97: Evaluate: $|-2+5i|$

$$|-2+5i| = \sqrt{(-2)^2 + (5)^2} = \sqrt{4+25} = \sqrt{29}$$

Q99: Solve using quadratic formula.

$x^2 + 6x + 3 = 0$, a=1, b=6, c=3

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-6 \pm \sqrt{6^2 - 4(1)(3)}}{2(1)} = \frac{-6 \pm \sqrt{24}}{2} = -3 \pm \sqrt{6}$$

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Q: 100: - A sample of size n=3 drawn without replacement experiment from a population N=5 items whose values 0, 2, 3, 6, 7 draw possible samples

Sample without replacement = ${}^{N}C_{n}=5C3=10$

(0,2,3), (0,2,6), (0,2,7), (0,3,6), (0,3,7), (0,6,7), (2,3,6), (2,3,7), (2,6,7), (3,6,7).

Q101: Describe about significance level.

The significance level is the criterion used for rejecting the null hypothesis. It show you how likely a result is due to chance. The most frequently used values of α, the significance level, are 0.05 and 0.01, i.e. 5 percent and 1 percent.

Q102: Which of the following statement represents continuous data and discrete data? (5)

i.
ii. Number of shops in a plaza. **Discrete data**
iii. Hourly temperature recorded by whether bureau. **Continuous data**
iv. Inches of rainfall in a city. **Continuous data**
v. Number of passengers carried by rail every year. **Discrete data**
vi. Height measurements of boys studying in a college. **Discrete data**

Q103: Complete the ANOVA Table

Source of variation	Sum of squares	Degree of freedom	Mean Square	F
Between treatments		4	19.86	6.9
Error				
Total	137.04	24		

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Solution:

Source of variation	Sum of squares	Degree of freedom	Mean Square	F
Between treatments	79.44	4	19.86	6.9
Error	57.6	20	2.88	
Total	137.04	24		

Explanation:

Source of variation	Sum of squares	Degree of freedom	Mean Square	F
Between treatments	19.85x4=79.44	4	19.86	6.9
Error	137.04-79.44= 57.6	24-4=20	57.8/20=2.88	

Total	137.04	24		

Q104:Let T1 and T2 are two unbiased estimators. The variances of both estimators are given below: $V(\mathrm{T}_1)=\frac{\sigma^2}{3}$ *and* $V(T_2)=\frac{3\sigma^2}{2}$

Solution:

As 1/3 < 3/2

Hence Var (T1) < Var (T2),

Therefore, T1 is more efficient than T2

Efficiency is given below

$$E_f=\frac{V(\mathrm{T}_2)}{V(T_1)}$$

$$=\frac{\frac{3\sigma^2}{2}}{\frac{\sigma^2}{3}}=\frac{3\sigma^2}{2}\times\frac{3}{\sigma^2}=\frac{9}{2}$$

Q105: Find value of sin 120 without using calculator

Sin (120) =Sin (180-120) =sin (60) = $\frac{\sqrt{3}}{2}$

Q106: Find value of COS 135 without using calculator

COS (135) = Cos (180-45) = -Cos45= $-\frac{1}{\sqrt{2}}$

Note: to solve without calculator question use following table

	30^0	45^0	60^0
Sin	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$
Cos	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$
Tan	$\frac{1}{\sqrt{3}}$	**1**	$\sqrt{3}$

Q107: There are two populations A and B, have means 8 and 7 respectively. Calculate the mean sampling distribution of $\bar{X}_A-\bar{X}_B$

The mean of sampling distribution of $\bar{X}_A-\bar{X}_B$ is denoted by $\mu_{\bar{X}_A-\bar{X}_B}$ is equal to the difference between populations mean that is

$\mu_A-\mu_B$ =8-7=1

$\mu_{\bar{X}_A-\bar{X}_B}=\mu_A-\mu_B=1$

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Q108: For given values, calculate Pearson coefficient of skewness using mode of the data.

The Standard deviation is 1.36. Data: 5,3,4,4,2,1

Mode=4

Mean= $\frac{\sum X}{n} = \frac{19}{5} = 3.8$

$$Coefficien\ of\ skewness = \frac{mean - \text{mod}\, e}{S.D} = \frac{3.8-4}{1.36} = -0.147$$

Q109: Random samples of size n=2 is drawn with replacement from population consisting of three number 0, 2, 5. Draw all possible samples and sample mean.

Sample	Mean
0,0	0+0/2=0
0,2	0+2/2=1
0,5	2.5
2,0	1
2,2	2
2,5	3.5
5,0	2.5
5,2	3.5
5,5	5

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Q110: Why t distribution is whole family of distribution?

The number of degrees of freedom determines the shape of the t-distribution.Thus there is a different t-distribution for each number of degrees of freedom. That's why, it is called a whole family of distributions.

Q111: The 1st and 3rd quartile of a data are Q1=67, and Q3=83. Find the values of mid-quartile range.

$$Mid - quartile\,range = \frac{Q_1 + Q_3}{2}$$

$$Mid - quartile\,range = \frac{67+83}{2} = \frac{150}{2} = 75$$

Q112: Evaluate $\frac{2+6i}{3+5i}$

$$\frac{2+6i}{3+5i} = \frac{2+6i}{3+5i} \times \frac{3-5i}{3-5i}$$
$$= \frac{(2+6i)(3-5i)}{(3)^2-(5i)^2} = \frac{6-10i+18i-30i^2}{9-25i^2} = \frac{6+8i-30(-1)}{9-25(-1)}$$
$$= \frac{6+8i+30}{9+25} = \frac{36+8i}{34}$$

Q113: Write disadvantages of range.

1. It ignores all the INFORMATION available from the intermediate observations.

2. It might give a MISLEADING picture of the spread in the data.

Q114: Write properties of echelon form of matrix.

A matrix is in echelon form if it has the following properties

- Every non-zero row begins with a 1 (called a leading 1)
- Every leading one in a lower row is further to the right of the leading one above it.
- If there are zero rows, they are at the end of the matrix

Q115: Write names of different types of correlation in graph?

- Positive linear correlation
- Perfect Positive Linear Correlation
- Completely Random
- Extreme Of Dissociation

Q116: Find the first two expressions of $(2-x)^6$ in as ascending power of x.

First expression: $(2-x)^1=2-x$

Second Expression: $(2-x)^1=4-4x-x^2$

Q117: Find out mean deviation and $(X-\bar{X})=+3,-3,-4,-1,+1,+4$.

$$\left|X-\bar{X}\right|=+3,+3,+4,+1,+1,+4$$

$$M.D=\frac{\sum\left|X-\bar{X}\right|}{n}=\frac{3+3+4+1+1+4}{6}=\frac{16}{6}=2.67$$

Q118: A random variable X is normally distributed with $\mu=50$ and $\sigma^2=25.5$ find the probability of X larger than 54.

Applying z-test

$$z=\frac{x-\mu}{\sqrt{\sigma^2}}=\frac{54-50}{\sqrt{25.5}}=\frac{-4}{5.05}=-1.07$$

Now we need to find the area under the normal curve from z= -∞ to Z = -1.07. Looking at

The area table of the standard normal distribution, we find that Area from 0 to 1.07=0.3577

P(X>54) = P(Z>-1.07)=0.5-0.3577=0.1423

Q119: $\cos\theta=\frac{1}{2}$, find all values of θ giving all the roots in $(0\le\theta\le360)$.

$$\cos^{-1}(\frac{1}{2})=60$$

One solution is 60

To find all intervals we use periodic property of cos

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$$\cos(60-360)=\cos(-300)=\cos(300)$$

Q120: $\frac{10}{1} and \frac{21}{7}$ **are rational or irrational numbers.**

Both are rational numbers because they can be written in p/q form

Q121: How many 2 permutations for {A, B, C}

3P2= $= \frac{3!}{(3-2)!} = 3! = 3.2.1 = 6$

Q122: Write the geometric sequence formula

$a_n = ar^{n-1}$

Q123: If A={1,2,3}, B={1,3,5} and C={2,4,6} then show that (A ∪ B) ∪ C = A ∪ (B ∪ C).

(A ∪ B) ∪ C = A ∪ (B ∪ C)

(A ∪ B) ∪ C = ({1,2,3} U {1,3,5}) U {2,4,6}

L.H.S= ={1,2,3,5} U {2,4,6}= {1,2,3,4,5,6}

R.H.S= A ∪ (B ∪ C) = {1,2,3} U ({1,3,5} U {2,4,6})

={1,2,3} U {1,2,3,4, 5,6}= {1,2,3,4,5,6}

Q124: If the function f is defined by $f(x)=\frac{3}{x}, \quad 0 < x \leq 3$ **Then find the domain and range of f.**

Domain of X=R-{3}

Range of X = R

Q125: Write properties of reduced echelon form of matrix.

- Every non-zero row begins with a 1 (called a leading 1)
- Every leading one in a lower row is further to the right of the leading one above it.
- If there are zero rows, they are at the end of the matrix

Q126: Suppose that X={ a,b } are two possible values for combination of X. Find the value of C(2,2).

C(2,2)= $\frac{n!}{r!(n-r)!} = \frac{2!}{2!} = 1$

Q127: If A={2,4,6,8,10,12,14,19} and B={12,13,14,15,16} then find AUB.

AUB={2,4,6,8,10,12,14,19} U {12,13,14,15,16}

AUB= {2,4,6,8,10,12,13,14,15,16,19}

Q128: Let f(x) =5x^3+3x-9 and g(x)=x^2 then find fog(x).

fog(x)= f(g(x)) = 5(x^2)^3+3(x^2)-9 = 5x^6+3x^2-9

Q129: What will be the probability of one card out of 52 playing cards?

P (one card) =1/52

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Q130: For the normal distribution with $\mu = 55$ and $\sigma = 10$. How much area will be found under the curve?

Applying z-test

$$z = \frac{x-\mu}{\sigma} = \frac{1-55}{10} = \frac{-54}{10} = -5.4$$

Now we need to find the area under the normal curve from z= -∞ to Z = -2.5. Looking at

The area table of the standard normal distribution, we find that Area from 0 to 2.5 =

0.4938

So required area is 0.5-0.3577=0.1423

Q131: If A={0,1,2,…,100} and B={0,1,2,…,1000} write them in set builder notation.

A={x | x < 100 and x is a whole number}

B= {x | x < 6 and x is a whole number}

Q132: Which Scale of measurement is used in

- **Temperature** **(Interval Scale)**
- **Weight** **(Ratio Scale)**
- **Money** **(Ratio Scale)**
- **Marital status** **(Nominal Scale**
- **Performance of student** **(Ordinal Scale)**

Q133: If $A = \begin{bmatrix} 0 & -1 \\ 1 & 3 \end{bmatrix}, B = \begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix}$ then prove that 3B-3A=-3(A-B)

3B-3A= $3\begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix} - 3\begin{bmatrix} 0 & -1 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 9 & 6 \\ 6 & 3 \end{bmatrix} - \begin{bmatrix} 0 & -3 \\ 3 & 9 \end{bmatrix} = \begin{bmatrix} 9-0 & 6-(-3) \\ 6-3 & 3-9 \end{bmatrix} = \begin{bmatrix} 9 & 9 \\ 3 & -6 \end{bmatrix}$

-3(A-B)= $-3\left(\begin{bmatrix} 0 & -1 \\ 1 & 3 \end{bmatrix} - \begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix}\right) = -3\begin{bmatrix} 0-3 & -1-2 \\ 1-2 & 3-1 \end{bmatrix} = -3\begin{bmatrix} -3 & -3 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 9 & 9 \\ 3 & -6 \end{bmatrix}$

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https://www.youtube.com/channel/UCfQB0nFXMlE9EuCAWYoUGTA